

SPEAKER BRIEFING · SMART INDIA HACKATHON

What's actually running inside POLAR-EMS

A working reference for presenting the live demo: what the simulation is really computing, what every number on screen means, how the two control algorithms differ, and how this maps onto a genuine polar-station engineering problem.

SIMULATION

100% client-side

CONTROLLERS

2 (baseline + AI)

PRIORITY TIERS

4

DEMO SCENARIOS

8

LIVE PAGES

7

01 What the website is actually doing

It is not a slideshow of pretty numbers. It is a small, running physics-and-rules model of a diesel/wind/solar/battery microgrid, ticking forward in simulated time, with two independent control algorithms competing to serve the same load from the same weather.

Every second of real time, the page advances the simulated clock by a few hours (configurable 1×/4×/12×) and re-runs the full pipeline: generate this tick's weather → compute renewable output → compute station load → run the rule-based baseline controller → run the AI-heuristic optimizer → apply the hard safety layer → check for anomalies → update forecasts → accumulate the green-audit totals → redraw whichever tab is open. Nothing is pre-recorded; every chart you point at during the demo is being generated live in the browser as you talk.

WHY THIS MATTERS FOR JUDGES

If someone asks "is this just a mockup with fake numbers," the honest and correct answer is: no — it's a deterministic simulation with real physical relationships (power curves, seasonal solar cycles, battery energy balance, fuel-burn curves), and every KPI is derived from that running state, not hardcoded.

The one sentence to open with

"Diesel is scarce and expensive, the internet can vanish for weeks, and losing power at -45°C is a safety event — so POLAR-EMS forecasts load and renewables, optimizes the diesel/battery/wind/solar mix in real time, and keeps making safe decisions with zero cloud connectivity."

02 The five layers, and where the demo lives

The whole point of the architecture is that each layer can keep working if the one above it disappears. The live page demonstrates layers 1–4 running entirely in-browser; layer 5 (cloud) is simulated by the Offline Autonomy tab.

LAYER	WHAT IT DOES HERE	SURVIVES
1• Field sensors	Weather, generation, load and battery/diesel telemetry — produced by the physical model each tick	Total isolation
2• Edge gateway	Ingests that telemetry and runs every model below, entirely client-side	Zero internet
3• AI engine	Load/renewable forecasting, dispatch optimization, anomaly detection	Sparse data
4• Dispatch & safety	Executes the chosen controller's decision; hard safety rules can override it	AI/model failure
5• Cloud portal	Represented by the sync outbox on the Offline Autonomy tab	Days-to-weeks offline

03 Every parameter, explained

Grouped exactly as they appear across the seven tabs. Use this as the glossary if a judge points at a number and asks "what's that."

Weather & season model

PARAMETER	MEANING	HOW IT'S GENERATED
Temperature	Outside air temp, $^{\circ}\text{C}$	Seasonal sine curve (-45°C deep winter to -5°C deep summer) + daily swing + noise

PARAMETER	MEANING	HOW IT'S GENERATED
Wind speed	m/s at hub height	Smoothed random walk (autocorrelated) around a seasonal mean, so it drifts realistically instead of jittering tick to tick
Irradiance	Solar input, W/m ²	Seasonal sun potential × time-of-day factor × cloud cover, forced to zero through the modeled polar-night window
Season label	Polar night / winter / summer / midnight sun	Derived from the same seasonal function that drives solar potential

Renewable generation

PARAMETER	MEANING	HOW IT'S GENERATED
Wind output	kW from the turbine array	Standard turbine power curve: 0 below 3.5 m/s cut-in, cubic ramp to rated output at 12 m/s, 0 above 25 m/s cut-out
Solar output	kW from the PV array	$\text{Irradiance} \div 1000 \times \text{array capacity} \times 92\% \text{ system efficiency}$
Renewable fraction	% of load served without diesel	Renewables actually absorbed by load, divided by total load, per tick and cumulatively

Station load — four priority tiers

This hierarchy is enforced everywhere shedding happens: amenities go first, then general power, then research, and life-support/heating are only ever touched by the hard safety override, never by an ordinary dispatch decision.

TIER	LOAD	TYPICAL SHARE	SHED ORDER
P1	Life support + heating (heating rises as temperature falls)	~40–45%	Never, except critical-SOC override
P2	Research / lab equipment	~20–25%	Third
P3	General station power	~15–20%	Second
P4	Amenities	~8–10%	First

Battery

PARAMETER	MEANING
SOC	State of charge, % of 500 kWh usable capacity
Charge / discharge rate	Capped at 150 kW in either direction
Reserve floor	Baseline controller: fixed 20%. AI controller: 20–32%, raised when the 3-hour renewable forecast is weak
Critical floor	10% — below this the hard safety layer takes over regardless of which controller is active
Temperature / cycles / health	Tracked for the anomaly & maintenance page; excess cycling and thermal drift degrade the health score

Diesel generator

PARAMETER	MEANING
Rated output	300 kW
Specific fuel consumption (SFC)	Litres of diesel per kWh delivered — base 0.30 L/kWh, with a penalty modeled when the generator runs below ~50% load (real diesel gensets are least efficient at low load, which is exactly why the AI controller avoids starting it for small shortfalls)
Fuel tank	12,000 L capacity
Runtime hours	Cumulative hours running — the number the optimizer is explicitly trying to minimize

Forecast metrics (Forecast tab)

PARAMETER	MEANING
Horizon	How far ahead the prediction targets: 1h, 6h, 12h or 24h
MAE	Mean absolute error between predicted and realized load, in kW — "on average, how far off"
RMSE	Root-mean-square error — same units, penalizes large misses more than MAE
Samples scored	How many predictions have actually been resolved against reality so far in this run

Optimizer / dispatch outputs (Optimizer tab)

PARAMETER	MEANING
Renewables served	kW of the current load coming directly from wind + solar
Battery	kW flowing into (+) or out of (–) the battery this tick
Diesel	kW currently being delivered by the generator
Load shed	kW of non-critical load intentionally dropped (should be 0 in normal operation)
Why (reason)	A plain-English sentence the controller generates for its own decision every tick — this is the single most important feature for the "is this explainable" question

Anomalies & health (Anomalies tab)

PARAMETER	MEANING
Severity	Critical / warning / info
Detected vs. expected	The actual reading next to what the physical model says it should have been
Subsystem health (0–100%)	Battery, turbine array, generator — decays when anomalies fire, slowly recovers otherwise

Offline & sync (Offline Autonomy tab)

PARAMETER	MEANING
Outbox count	Telemetry records generated while offline, waiting to reach the cloud
Local subsystem status	Telemetry / AI forecasting / dispatch optimizer / safety controller — all shown ACTIVE independent of connectivity, which is the entire point of the tab

Green Audit (cumulative KPIs)

PARAMETER	MEANING
Diesel consumed / saved	Litres burned by the AI-run station vs. the shadow rule-based baseline, since the run started
CO ₂ avoided	Fuel saved × 2.68 kg CO ₂ per litre (standard diesel emission factor)
Renewable fraction	Share of all energy served by wind + solar over the whole run
Runtime reduction	Generator hours saved vs. baseline

PARAMETER	MEANING
What-if projection	A capacity-ratio scaling of the currently measured fuel rate — explicitly labeled a projection, not a re-simulation

04 How the internals actually work

This is the part that answers "so what's the AI actually doing" without hand-waving.

Two controllers, same inputs, every tick

Both controllers see the identical weather draw and identical load for that tick, so any difference in outcome is attributable to the control logic itself — not to lucky weather. This is what makes the fuel-savings number on the Optimizer and Green Audit tabs a measured result rather than an assumption.

● Rule-based baseline

Logic

- Serve load from renewables first, always
- Any surplus charges the battery; any deficit discharges it
- Fixed 20% reserve floor, no matter the weather outlook
- If the battery can't cover the rest, start diesel for exactly the shortfall
- No forecast awareness — purely reactive to the current tick

● POLAR-EMS optimizer

Logic

- Same renewables-first, battery-second waterfall
- Reserve floor moves to 32% when the next-3-hour renewable forecast is weak, 20% when it's strong — it holds back charge when it expects to need it
- When diesel must start, it deliberately runs a little above the immediate shortfall to also recharge the battery, so it doesn't have to cold-start again soon
- Diesel is avoided specifically because it's least fuel-efficient at low partial load — the SFC curve models this directly

THE JUDGE-FACING PUNCHLINE

The baseline is not a strawman — it's a fully functional, sensible controller. The optimizer beats it only because it looks ahead and times diesel starts smarter. That's the actual value proposition of

"AI-driven": not that rules are broken, but that a forecast-aware policy burns less fuel and runs the generator fewer hours for the identical job.

Hard safety layer — runs after both controllers, every tick

1 Critical SOC override

If battery SOC drops below 10% regardless of which controller is driving, diesel is forced to full rated output and non-critical tiers are shed — this check cannot be disabled by either controller.

2 AI/sensor failure override

Triggering the "AI service failure" scenario doesn't just change a label — it actually re-routes the live dispatch through the rule-based controller's logic for as long as the fault is active. The mode badge (

● AI OPTIMIZED

/ ● SAFE FALLBACK

) always reflects which logic is actually driving the station, not a cosmetic state.

Anomaly detection — statistical, not a black box

Every anomaly check compares an actual reading against what the physical model says it should be, and only fires when the gap crosses a threshold. That's deliberate: a judge can ask "how do you know it's anomalous" and the answer is always a specific expected-vs-detected comparison, never "the model said so."

CHECK	TRIGGER
Turbine underperformance	Actual wind output >25% below the power-curve prediction for the current wind speed
Battery thermal / degradation	Temperature drifting outside -10°C to 15°C with an accelerated rise
Generator efficiency drift	Specific fuel consumption >15% above the baseline value
Sensor fault	An implausible reading (e.g. a dropout to zero) inconsistent with surrounding telemetry

Forecasting — seasonal-naive, explicitly not a neural net

The forecast for a future hour blends the seasonal expectation for that hour (same time of year, same time of day) with the current deviation from that expectation, decayed the further out the prediction reaches:

```
predicted(t+h) = seasonal_baseline(t+h) + current_deviation * e-h/14 + noise
```

This is intentional: it's a simple, auditable statistical model rather than a trained network, which is easier to defend under judging time pressure and is exactly the kind of "explainable first" choice the safety framing calls for. Accuracy is verified, not assumed — every prediction is time-stamped, and once the simulation reaches that real moment, MAE/RMSE are recalculated against what actually happened.

Fuel & CO₂ accounting

```
fuel_used (L) = diesel_kw * SFC(load_fraction) * hours
fuel_saved = fuel_used_baseline - fuel_used_AI
CO2_avoided (kg) = fuel_saved * 2.68
```

2.68 kg CO₂ per litre is a standard published diesel combustion emission factor, not a tuned number — state that plainly if asked.

05 How this maps onto the real problem

Every feature in the demo exists because of a specific, named constraint of running a polar station — not because it looked good on a dashboard.

Fuel is scarce & costly

Every litre is shipped in at extreme cost. **Answer:** forecast-aware dispatch that avoids starting diesel for shortfalls the battery can cover, and avoids running it inefficiently at low partial load.

Internet is unreliable

Satellite links can drop for weeks. **Answer:** every control layer runs at the edge; the Offline Autonomy tab proves telemetry, forecasting, optimization and safety never pause, only the cloud sync queues.

Failure is a safety event

Losing heat at -45°C is life-threatening. **Answer:** a strict, independently-enforced load-priority hierarchy plus a hard-coded override that cannot be overruled by either controller.

Equipment degrades in extreme cold

Batteries, turbines and sensors drift under stress.

"Why should we trust an AI here?"

Answer: it's not opaque — every dispatch decision

Judges expect rigor, not vibes

Answer: the baseline vs. optimized comparison runs

Answer: statistical anomaly detection flags degradation early, with a plain-English recommendation, before it becomes a failure.

stores the reason it was made, and the safety layer sits outside the AI entirely, so trust doesn't depend on the model being right 100% of the time.

on identical conditions, so the fuel/CO₂ savings shown are a measured delta, not an assumed one — and change if the scenario changes.

06 Judge Q&A cheat sheet

Straight answers, no overclaiming — this is the tone that reads as engineering maturity rather than a pitch.

Is this real station data?

No, and the site says so explicitly. It's a synthetic physical model (power curves, seasonal solar/polar-night cycles, priority-weighted load, battery/diesel dynamics) — not random numbers, and not a live feed. That's a deliberate, stated engineering choice for a hackathon build.

Why AI/optimization instead of rules only?

Rules alone can't adapt their diesel/battery balance to a shifting weather forecast. The demo proves the delta directly: identical conditions, two controllers, one measurably burns less fuel and runs the generator fewer hours.

Why not deep learning / RL for dispatch?

A constrained, rule-explainable heuristic is verifiable and safe to reason about for a system where a wrong decision has safety consequences. That matters more here than raw model sophistication.

What happens if the AI is just wrong, or unavailable?

The hard safety layer sits outside the optimizer and can't be overridden by it. Trigger the "AI service failure" scenario live — the dashboard visibly switches to SAFE FALLBACK and dispatch keeps running on deterministic rules.

What happens with zero internet for weeks?

Nothing changes operationally — telemetry, forecasting, optimization and safety all run at the edge. Only the cloud sync queues records in a local outbox until connectivity returns, then uploads in a batch.

How is fuel/CO₂ savings calculated?

Two controllers run in parallel on identical weather and load. Fuel saved is the litre difference between them; CO₂ avoided is that difference times the standard 2.68 kg/L diesel emission factor.

What would it take to run this on a real station?

Swap the synthetic telemetry generator for real sensor/BMS/genset feeds into the same edge pipeline — the forecasting, optimization, safety and anomaly-detection logic are all designed to consume exactly that shape of data already.